DOOR STATUS INDICATOR MODULE

PCB Design using KiCad — Project Report

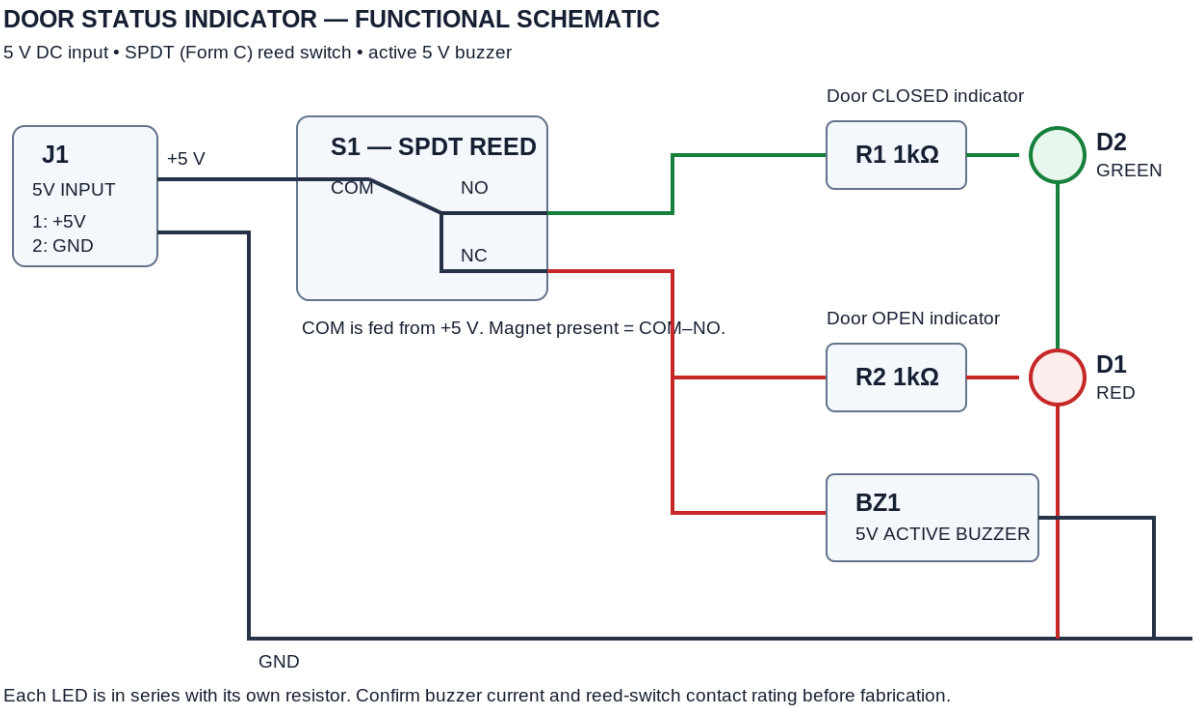
Student Name	Lakshya
Course / Branch	ECE
Subject	PCB Design
Project Title	Door Status Indicator Module

Prepared for the individual PCB design assignment.

1. Project Overview

The module indicates whether a door is open or closed. A magnet is mounted on the moving door and an SPDT magnetic reed switch is mounted on the fixed frame. When the door is closed, the magnet actuates COM–NO and the green LED turns on. When the door opens, COM–NC powers the red LED and active buzzer.

2. Circuit Schematic



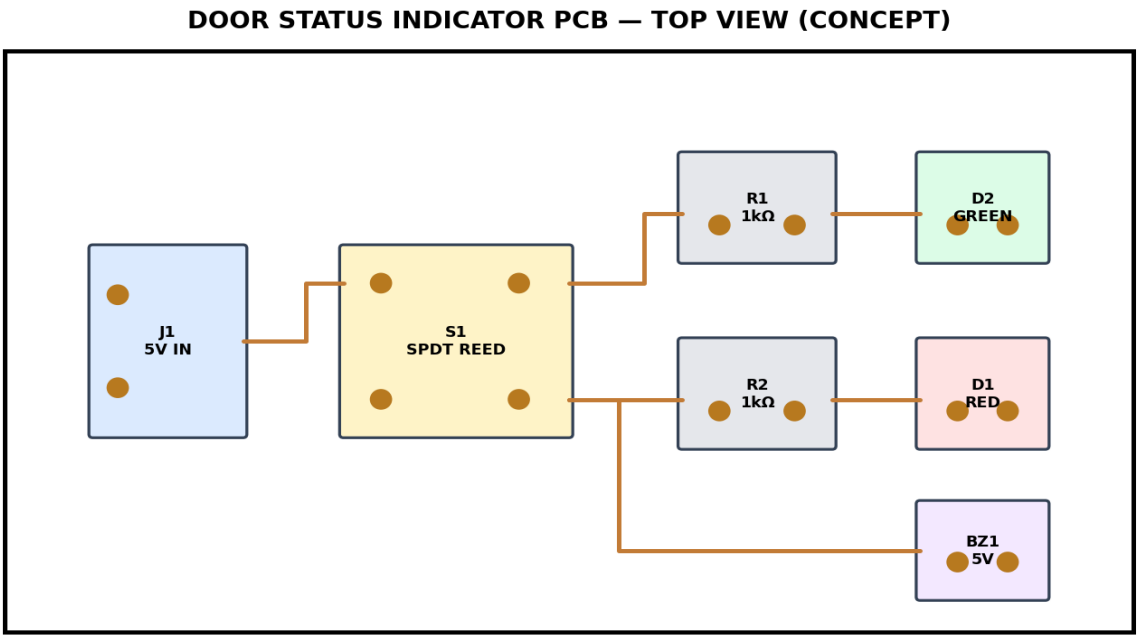
Circuit assumption: a 3-terminal SPDT (Form C) reed switch is used. A 2-terminal reed switch is not a direct replacement. Use an active 5 V buzzer; check its current rating against the reed contact rating and supply capacity.

3. Footprint Assignment

These are starting selections only. Verify actual body dimensions, pin spacing, polarity and mounting style against the exact parts purchased. The reed switch and buzzer footprints are component-specific.

Ref	Component	Suggested footprint
J1	2-pin 5 V power connector	2-pin terminal block, 5.08 mm pitch
S1	SPDT reed switch	Custom 3-pin footprint matching datasheet
D1, D2	5 mm LEDs	LED_THT:LED_D5.0mm
R1, R2	1 kΩ, 1/4 W resistors	DIN0207 axial, 7.62 mm pitch
BZ1	5 V active buzzer	Through-hole footprint matching datasheet

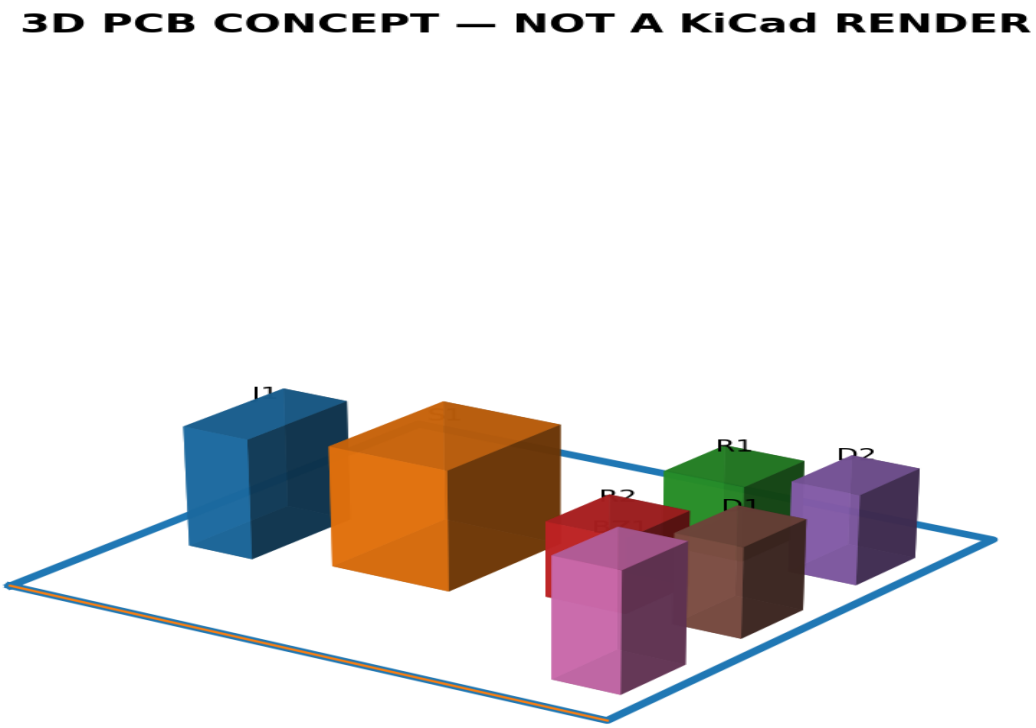
4. PCB Layout — 2D Top View



Illustrative placement/routing only — verify and capture final screenshots in KiCad

Placement intent: J1 near a board edge; sensor wiring routed cleanly; each LED grouped with its series resistor; buzzer near an accessible edge. Maintain board-edge and copper clearances.

5. PCB Layout — 3D View



This is a conceptual illustration, not a KiCad 3D Viewer screenshot. Open the completed PCB in KiCad 3D Viewer to capture the required genuine image.

6. Bill of Materials (BOM)

Ref	Qty	Component	Suggested footprint / package	Value / rating
J1	1	2-pin power connector	TerminalBlock_Phoenix: terminal block 2-pin 5.08 mm	5 V DC input
S1	1	SPDT magnetic reed switch	Through-hole 3-pin reed switch footprint (match purchased part)	Form C, magnet-actuated
D1	1	Red LED	LED_THT:LED_D5.0mm	Red, 3 mm or 5 mm
D2	1	Green LED	LED_THT:LED_D5.0mm	Green, 3 mm or 5 mm
R1	1	Resistor	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	2 kΩ, 1/4 W
R2	1	Resistor	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	2 kΩ, 1/4 W
BZ1	1	Active buzzer	Buzzer_Beeper: Buzzer_12x9.5RM7.6	5 V active, polarity marked

7. Final Verification and Submission Checklist

In KiCad: create and annotate the schematic; run ERC; assign footprints matching datasheets; update PCB from schematic; place and route all components; run DRC and resolve errors; inspect the 3D board; capture genuine screenshots of the routed board, 2D PCB and 3D PCB; plot Gerber copper, solder mask, silkscreen and Edge.Cuts layers; generate drill files. Upload the report and Gerber/drill files to a personal GitHub repository. Create a Word document containing the student name and GitHub URL, export it as PDF, and submit that PDF to LMS.

Important: The included 2D/3D views are design illustrations, not KiCad screenshots. This package does not contain verified Gerber files. Do not submit the illustrations as evidence of a routed KiCad board.

Appendix: BOM.csv and design illustration source files are included in the accompanying ZIP package.